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Multi-Timescale and Multi-Agent Collaborative Allocation of Emergency Resources for Natural Disasters

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Abstract: The allocation of emergency resources serves as a critical lifeline in natural disaster relief, fundamentally determining survival rates, resources rescue efficacy, and the restoration of social order. There are two critical limitations in existing allocation models: 1) inadequate cross-agent coordination regarding fund and resource flows; 2) insufficient adaptability to dynamic uncertainties. To address these gaps, this study first proposes and models a multi-agent linkage mechanism to synchronize decision agent (fund), execution agent (procurement/transportation), and demand agent in disaster areas to holistically optimize economic costs, life-saving utility, and supply-demand balance. Multi-stage adaptive robust optimization is adopted to transform the above allocation model to a multi-timescale feedback model, integrating pre-disaster planning and post-disaster periodic adjustments under demand, procurement, and transportation uncertainties. For the convenience of solving, a linear reformulation technique is developed to convert the aforementioned nonlinear robust model into a tractable linear program using duality theory and semi-infinite constraint handling. A case study based on an earthquake scenario demonstrates the multi-timescale and multi-agent collaborative mechanisms significantly enhance resource allocation optimality, stability and robustness under multiple sources of uncertainty. Sensitivity analyses reveal optimal inflection points for key parameters such as time discretization and historical dependency depth, providing practical guidance for model deployment. Compared to deterministic, static robust, and rolling horizon methods, this study realizes notable improvements in allocation safety and a comprehensive rescue objective.

Keywords: Emergency resource allocation; multi-agent collaboration; multi-timescale adjustment; uncertainty; natural disaster

1 Introduction

Recent acceleration in global climate change and urbanization has precipitated frequent natural disasters (e.g., earthquakes, floods, typhoons) and public emergencies (e.g., pandemics, safety incidents) [1]. The IPCC AR6 report indicates that the probability of disasters such as heatwaves, torrential rains, and floods has surged over fivefold compared to pre-industrial levels [2]. In Q1 2025 alone, direct economic losses from natural disasters in China reached ¥10.16 billion, with compound disasters (e.g., earthquake-triggered landslides, typhoons compounded by flooding) accounting for over 40% [3]. These events pose severe threats to socioeconomic stability and public safety. Crucially, disaster-induced losses stem not only from the events themselves but also from inefficient emergency response—delayed decisions or suboptimal resource allocation significantly undermine rescue efficacy, exacerbating human suffering and economic damage. During the post-disaster "golden 72-hour window", survivor survival rates decline precipitously—from ~90% within the first 24 hours to 20–30% after 72 hours [4]. This imposes extreme demands on the timeliness, coordination, and robustness of emergency resource allocation. The effective allocation of emergency resources is critically important for minimizing casualties, mitigating property losses, and alleviating human suffering in the aftermath of disasters [5].

As a core component of disaster management, post-disaster resource allocation has long been prioritized in academia and practice. Extensive studies have systematically advanced theoretical and methodological frameworks, spanning emergency logistics design, resource optimization, and intelligent scheduling technologies [6-8]. Traditional models emphasize facility location, routing optimization, and efficient scheduling to enhance response speed and material guarantee capacity [7,9,10]. For instance, Wang [7] proposed a multi-stage location-routing model for earthquake rescue, while Bi and Zhang [6] systematized emergency logistics for bulk commodities like grain. Scenario-specific models (e.g., forest fires, floods) have also been developed, such as multi-depot resource scheduling framework [9] and reservoir flood-control system [11], effectively reducing resource consumption and disaster impacts.

Recent urbanization, climate change, and compound disasters have exponentially increased the complexity

of post-disaster response, exposing limitations in single-agent and static optimization approaches [8,12]. Multi-agent collaborative scheduling has thus emerged as a critical research frontier. Yi and Ozdamar [10] pioneered an integrated logistics model coordinating location-distribution with evacuation, highlighting synergy between temporary emergency centers and medical resources. Procter, et al. [13] demonstrated through rockfall case studies that multidisciplinary, multi-skilled team collaboration is essential for complex search-and-rescue operations. Ren and Zhang [14] further optimized volunteer dispatch using multi-objective scheduling incorporating fatigue effects and survivor satisfaction, proposing multi-agent algorithms to enhance efficiency and fairness. Wu, et al. [15] established a dynamic game-theoretic model for cross-regional joint dispatch in the Guangdong-Hong Kong-Macao Greater Bay Area, simulating interactions between resource managers and disaster dynamics to address compound events.

Addressing demand uncertainty and dynamic disaster evolution remains a core challenge. Dominant methodologies include rolling optimization, robust optimization, dynamic games, and reinforcement learning [15-17]. Wu, et al. [16] integrated genetic algorithms with reinforcement learning to achieve intelligent, adaptive allocation of maritime rescue resources, markedly improving decision efficiency and stability in volatile environments. Zheng [17] proposed a multi-stage economic scheduling model with uncertainty quantification for post-disaster energy systems using mobile electric vehicles, enabling flexible supply amid infrastructure damage. Li, et al. [12] developed a two-stage material distribution framework for urban rainstorm floods, employing multi-objective optimization to balance timeliness, equity, and social costs, while leveraging rolling data and Pareto solutions to enhance dynamic adaptability. Yang, et al. [18] introduced a demand-urgency-based vehicle routing model with road condition adjustments, substantially reducing system losses and resource shortage risks.

Moreover, information integration and intelligent scheduling technologies have diversified implementation pathways. Data fusion techniques predict post-disaster disease spectra and medical needs [19], while multi-carrier drone communication systems improve data acquisition and survivor tracking in compromised areas [20]. Cheng, et al. [21] designed a hybrid genetic-large neighborhood search algorithm for multi-type helicopter scheduling, overcoming local optima in complex missions. Tian, et al. [22] developed high-efficiency routing algorithms for special vehicles in large-scale road networks, bolstering computational support for cross-regional dispatch. Yang, et al. [23] emphasized terminal command platforms integrating "last-mile" delivery with on-site coordination, enhancing flexibility and precision. Liu, et al. [24] validated through large-scale medical data that multi-department coordination significantly improves epidemic control and psychological intervention. Davis, et al. [11] revealed post-hurricane spikes in specific emergency needs (e.g., respiratory distress, seizures) using 911 call data, underscoring the need for targeted resource reconfiguration.

Regarding the allocation optimization of emergency resources after natural disasters, despite advances in multi-agent collaboration, dynamic scheduling, and intelligent optimization, critical gaps persist: **Limited cross-agent coordination**: Most models focus on single-agent resource/fund allocation, failing to synchronize funding allocation, resource procurement, transportation, and demand. This impedes information feedback and resource handover efficiency. **Inadequate uncertainty adaptation**: Robust and rolling optimization methods applied to emergency resource allocation lack multi-period feedback mechanisms, causing local optima and compromising economic efficiency and response effectiveness. **Computational intractability**: Complex nonlinear models exhibit low solving efficiency for large-scale scenarios, hindering practical deployment. To bridge these gaps, we propose a novel tripartite "Decision-Execution-Demand" collaborative mechanism for emergency resource allocation, integrating multi-stage adaptive robust optimization and linearization techniques to enhance adaptability, solvability, and applicability. Key innovations and contributions include:

- **Multi-agent collaborative allocation model:** Through bidirectional information feedback channels, we establish a collaborative allocation model that integrates decision, execution, and demand agents to efficiently synchronize funding allocation, resource procurement, transportation, and demand. This framework achieves comprehensive optimization of three objectives: economic costs of resource allocation, post-disaster life-saving utility, and supply-demand gap of emergency resources (Section 3).
- **Multi-timescale feedback adjustment model:** Using multi-stage adaptive robust optimization, we incorporate multi-period uncertainty sets and feedback-based decision rules to formulate the above model as a dynamic allocation framework. This integrates pre-disaster planning with post-disaster multi-period adjustments, significantly enhancing both optimality and robustness of emergency resource allocation (Section 4).
- **Linear reformulation of allocation model:** Via linear programming duality principles and semi-infinite constraint techniques, the original nonlinear robust optimization model is converted into a computationally tractable linear programming formulation. This transformation substantially reduces solution complexity (Section 5).

2 Problem Description

2.1 Main Agents in Resource Allocation

According to the Emergency Response Law of the People's Republic of China [25], the national emergency management system emphasizes unified leadership, comprehensive coordination, hierarchical responsibility, and local management. The allocation of emergency resources represents a fundamental and critical challenge in the management of natural disaster emergencies [26]. Post-disaster emergency resource allocation involves multiple agents at various levels working in concert to facilitate disaster relief efforts [27]. Government agencies at all levels serve as the top-level decision agent, providing unified leadership for disaster response. They are responsible for formulating resource allocation plans, disbursing emergency relief funds, and ensuring the effective initiation of resource deployment activities. Their primary objective is to meet disaster relief needs while minimizing economic costs. Procurement and supply departments, acting as the intermediate execution agent, are responsible for procuring and transporting emergency resources at designated local transfer stations (TS). Their tasks are to ensure the timely preparation and accurate delivery of resources to the demand agent—maximizing the life-saving utility of the allocation. At the disaster areas (DA), survivors and disaster victims at affected sites are the demand agent. They require prompt access to emergency resources to maximize survival chances and aim to minimize the gap between resource supply and demand.

2.2 Collaborative Emergency Resource Allocation

The first 72 hours after a disaster are regarded as the "golden window" for emergency rescue. During this period, the decision, execution, and demand agents must collaborate closely to counteract the destructive impacts of natural disasters, as illustrated in **Figure 1**. Prior to or immediately after a natural disaster, the decision agent rapidly assesses the impact, and estimates the resource demand of affected populations (such as first-aid kits, instant food, drinking water, etc.). They issue allocation plans and disburse emergency funds to the execution agent, dynamically adjusting funding strategies based on real-time feedback regarding resource procurement and transportation. The execution agent implements the allocation plan by procuring, temporarily storing, and transporting emergency resources to the demand agent, adjusting procurement and transportation strategies in response to actual demands reported from the disaster areas. The demand agent—survivors at the disaster areas—cooperate actively with emergency response teams by reporting their actual demands and waiting in an orderly fashion for rescue and relief.

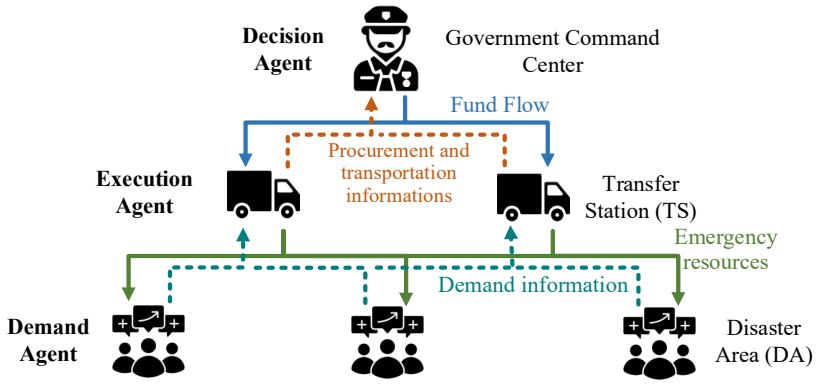

Figure 1 Schematic diagram of collaborative emergency resources allocation after natural disasters

3 Multi-Agent Collaborative Emergency Resource Allocation Model

Immediately before or after a natural disaster, a rapid assessment of the damage is conducted, and an emergency resource allocation plan is formulated for the critical 72-hour post-disaster period. For modeling convenience, this time horizon is discretized into T equal time periods (see **Figure 2**). A deterministic multi-agent collaborative resource allocation model is then developed, incorporating both constraints and optimization objectives.

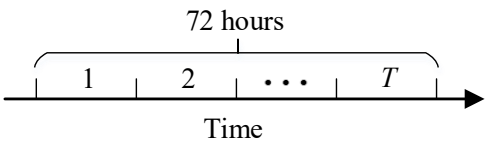

Figure 2 Multi-period discretization of 72-hour post-disaster period

Nomenclature

Index

t: Time period index, $\in\{1,\cdots,T\}$

i: Transfer station index, $\in\{1,\cdots,I\}$

j: Disaster area index, $\in\{1,\cdots,J\}$

k: Resource type index, $\in\{1,\cdots,K\}$

$\omega_1, \omega_2, \omega_3$: Weight coefficients for Z_1, Z_2, Z_3 in the objective function

Decision Variables

$Bud_{t,i}$: Funds allocated to TS i in period t

$Pur_{t,i,k}$: Quantity of resource k procured at TS i in

Parameter	period t
$Dem_{t,j,k}$: Demand for resource k at DA j in period t	$Tra_{t,i,j,k}$: Quantity of resource k transported from TS i to DA j in period t
Rat : Economic compensation ratio for transportation loss	**Auxiliary Variable**
$Uti_{t,j}$: Marginal life-saving utility at DA j in period t	$a_{t,j,k}$: Auxiliary variable for supply-demand gap at DA j for resource k in period t
$Cos_{t,i,j,k}$: Transportation cost of resource k from TS i to DA j in period t	**Objective Variable**
Pri_k: Unit purchase price of resource k	Z_1: Economic cost
	Z_2: Life-saving utility
	Z_3: Resource supply-demand gap
	f: Comprehensive optimization objective

3.1 Constraints

Fund Flow Constraint: For each transfer station in every period, the allocated fund must cover both the procurement and transportation costs, including additional expenses due to transportation losses (e.g., from damaged roads). Unused funds may be carried over to subsequent periods, but expenditures within each period must not exceed the cumulative fund allocation up to that period. Therefore, the funds allocated by the decision agent to the execution agent within a single period should satisfy Eq. (1).

$$\sum_{\tau=1}^{t}(Bud_{\tau,i} - \sum_{k=1}^{K} Pri_k \cdot Pur_{\tau,i,k} - \sum_{j=1}^{J}\sum_{k=1}^{K} Cos_{\tau,i,j,k} \cdot Tra_{\tau,i,j,k} \cdot (1 + Rat)) \geq 0 \; \forall i = 1, \cdots, I, t = 1, \cdots, T \tag{1}$$

Resource Flow Constraint: For each transfer station and resource type, surplus inventory after procurement and transportation in each period can be carried over. The quantity transported out up to each period must be less than the surplus inventory. Therefore, the procurement and transportation quantity of emergency resources by the execution agent should satisfy Eq. (2).

$$\sum_{\tau=1}^{t}(Pur_{\tau,i,k} - \sum_{j=1}^{J} Tra_{\tau,i,j,k}) \geq 0 \; \forall k = 1, \cdots, K, i = 1, \cdots, I, t = 1, \cdots, T \tag{2}$$

Supply-Demand Gap: For each disaster area and each resource type, the supply-demand gap $Res_{t,j,k}$ between execution and demand agents up to each period is accumulated with the time period, as constrained by Eq. (3). Here, $Res_{t,j,k}$ is only used as a symbolic representation and is not a model variable.

$$Res_{t,j,k} = \sum_{\tau=1}^{t}(\sum_{i=1}^{I} Tra_{\tau,i,j,k} - Dem_{\tau,j,k}) \; \forall k = 1, \cdots, K, j = 1, \cdots, J, t = 1, \cdots, T \tag{3}$$

Non-Negativity: All decision variables for fund, procurement, and transportation must be non-negative, i.e. Eq. (4).

$$Bud_{t,i} \geq 0, Pur_{t,i,k} \geq 0, Tra_{t,i,j,k} \geq 0 \; \forall k = 1, \cdots, K, j = 1, \cdots, J, i = 1, \cdots, I, t = 1, \cdots, T \tag{4}$$

3.2 Optimization Objective

Emergency resource allocation involves the optimization objectives of multiple agents. **Economic Cost Minimization** (Decision Agent): Minimize the total fund budget allocated over all periods and transfer stations, i.e. Eq. (5).

$$\min \; Z_1 = \sum_{t=1}^{T}\sum_{i=1}^{I} Bud_{t,i} \tag{5}$$

Life-Saving Utility Maximization (Execution Agent): Maximize the total life-saving utility, i.e., the sum of utility generated by resource deliveries across all periods, transfer stations, and disaster areas, as represented by Eq. (6).

$$\max \; Z_2 = \sum_{t=1}^{T}\sum_{i=1}^{I}\sum_{j=1}^{J} Uti_{t,j} \cdot Tra_{t,i,j,k} \tag{6}$$

Supply-Demand Gap Minimization (Demand Agent): To timely obtain emergency resources and avoid wasting resources as much as possible, minimize the total absolute supply-demand gap for all resources at all disaster areas and periods, as illustrated in Eq. (7). The absolute value operation introduces nonlinearity into the model. To facilitate solving the model, an auxiliary variable $a_{t,j,k}$ is introduced to represent the upper bound of $|Res_{t,j,k}|$. Eq. (7) is further transformed into linear expressions, namely Eq. (8) and Eq. (9).

$$\min \; Z_3 = \sum_{t=1}^{T}\sum_{j=1}^{J}\sum_{k=1}^{K} |Res_{t,j,k}| \tag{7}$$

$$\min \; Z_3 = \sum_{t=1}^{T}\sum_{j=1}^{J}\sum_{k=1}^{K} a_{t,j,k} \tag{8}$$

$$a_{t,j,k} \geq Res_{t,j,k}; \; a_{t,j,k} \geq -Res_{t,j,k} \; \forall k = 1, \cdots, K, j = 1, \cdots, J, t = 1, \cdots, T \tag{9}$$

Comprehensive Objective Minimization: A weighted sum of the above objectives is minimized to achieve a comprehensive optimization for all agents, i.e. Eq. (10).

$$\min \; f = \omega_1 \cdot Z_1 - \omega_2 \cdot Z_2 + \omega_3 \cdot Z_3 \tag{10}$$

4 Multi-Timescale Collaborative Resource Allocation Model

The deterministic allocation model described above relies on critical predefined parameters. However, these parameters are difficult to estimate accurately before or during a disaster, rendering the model's solution vulnerable to uncertainty and potentially infeasible. To dynamically adapt to uncertainty, we design a multi-timescale collaborative allocation strategy for emergency resources, integrating pre-disaster long-term planning with post-disaster periodic feedback adjustments, as illustrated in **Figure 3**. At the pre-disaster/initial

phase, based on initial parameter estimates from the execution agent, the decision agent formulates a resource allocation plan for the post-disaster "golden 72-hour window". For the post-disaster dynamic adjustment, as each time period elapses, demand uncertainties and disaster dynamics are progressively disclosed. At the start of each period, the decision and execution agents dynamically adjust the current allocation plan using feedback-based decision rules to address newly revealed uncertainties. This framework is formalized using multi-stage adaptive robust optimization (MSARO) method, involving periodic uncertain sets and recourse decision rules.

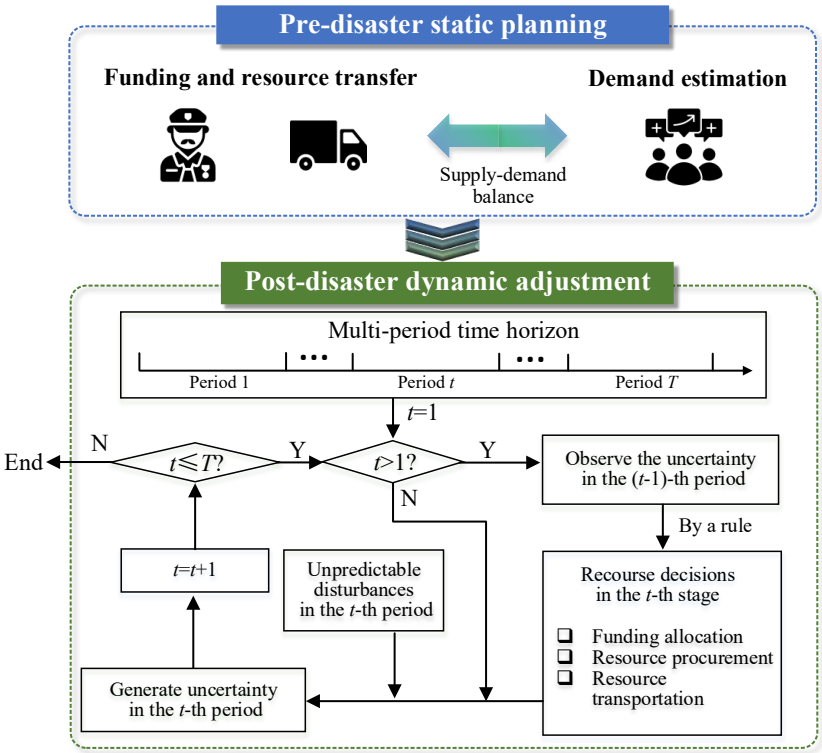

Figure 3 General idea of multi-timescale collaborative resource allocation

4.1 Multi-Period Uncertainty Sets

To reflect real-world disaster response, three key uncertainties affecting resource allocation are considered: For the resource demand uncertainty, actual demands at disaster areas are uncertain initially due to limited post-disaster information. Regarding resource procurement uncertainty, social constraints or disaster impacts may prevent the execution agent from procuring planned resources at transfer stations. Resource transportation uncertainty indicates that damaged infrastructure may impact planned deliveries from transfer stations to disaster areas.

Define several uncertain deviation parameters: $\xi_{t,j,k}^{\mathrm{Dem}}$ is uncertain demand deviation for resource k at DA j in period t; $\xi_{t,i,k}^{\mathrm{Pur}}$ indicates uncertain procurement deviation for resource k at TS i in period t; and $\xi_{t,i,j,k}^{\mathrm{Tra}}$ represents uncertain transportation deviation for resource k from TS i to DA j in period t. For compact representation, all $\xi_{t,j,k}^{\mathrm{Dem}}$s, $\xi_{t,i,k}^{\mathrm{Pur}}$s, and $\xi_{t,i,j,k}^{\mathrm{Tra}}$s in period t are separately represented as corresponding vector forms, i.e. $\boldsymbol{\xi}_t^{\mathrm{Dem}} = (\xi_{t,1,1}^{\mathrm{Dem}}, \cdots, \xi_{t,J,K}^{\mathrm{Dem}}) \in \mathbb{R}^{J\cdot K}$, $\boldsymbol{\xi}_t^{\mathrm{Pur}} = (\xi_{t,1,1}^{\mathrm{Pur}}, \cdots, \xi_{t,I,K}^{\mathrm{Pur}}) \in \mathbb{R}^{I\cdot K}$, $\boldsymbol{\xi}_t^{\mathrm{Tra}} = (\xi_{t,1,1,1}^{\mathrm{Tra}}, \cdots, \xi_{t,I,J,K}^{\mathrm{Tra}}) \in \mathbb{R}^{I\cdot J\cdot K}$. The actual values of resource demand and resource procurement and transportation can be further expressed as Eqs. (11)~(13), respectively. Here, $\widehat{Dem}_{\tau,j,k}$, $\widehat{Tra}_{t,i,j,k}$ and $\widehat{Pur}_{\tau,i,k}$ are separately the estimated values of $Dem_{t,j,k}$, $Tra_{t,i,j,k}$, and $Pur_{t,i,k}$.

$$Dem_{t,j,k} = \widehat{Dem}_{\tau,j,k} + \xi_{t,j,k}^{\mathrm{Dem}} \quad \forall k = 1,\cdots,K, j = 1,\cdots,J, t = 1,\cdots,T \tag{11}$$

$$Tra_{t,i,j,k} = \widehat{Tra}_{t,i,j,k} + \xi_{t,i,j,k}^{\mathrm{Tra}} \quad \forall k = 1,\cdots,K, i = 1,\cdots,I, j = 1,\cdots,J, t = 1,\cdots,T \tag{12}$$

$$Pur_{t,i,k} = \widehat{Pur}_{\tau,i,k} + \xi_{t,i,k}^{\mathrm{Pur}} \quad \forall k = 1,\cdots,K, i = 1,\cdots,I, t = 1,\cdots,T \tag{13}$$

Periodic uncertainty sets are essential for MSARO, as they must comprehensively characterize the key attributes of uncertain parameters within each time period. These sets enable the decision agent to better anticipate and respond to potential risks and variations, thereby ensuring the effective implementation of feedback adjustments. This study adopts box uncertainty sets [28] to model the uncertainty mentioned above. Specifically, box sets, i.e. Eqs. (14)~(16), define the bounded ranges of these parameters.

$$\Psi_{t,i,j,k}^{\mathrm{L,Tra}} \le \xi_{t,i,j,k}^{\mathrm{Tra}} \le \Psi_{t,i,j,k}^{\mathrm{U,Tra}} \ \forall k = 1,\cdots,K, i = 1,\cdots,I, j = 1,\cdots,J, t = 1,\cdots,T \tag{14}$$

$$\Psi_{t,j,k}^{\mathrm{L,Dem}} \le \xi_{t,j,k}^{\mathrm{Dem}} \le \Psi_{t,j,k}^{\mathrm{U,Dem}} \ \forall k = 1,\cdots,K, j = 1,\cdots,J, t = 1,\cdots,T \tag{15}$$

$$\Psi_{t,i,k}^{\mathrm{L,Pur}} \le \xi_{t,i,k}^{\mathrm{Pur}} \le \Psi_{t,i,k}^{\mathrm{U,Pur}} \ \forall k = 1, \cdots, K, i = 1, \cdots, I, t = 1, \cdots, T \tag{16}$$

For compact representation, the periodic uncertainty set in this study is expressed in vector-matrix form as Eq. (17), where: $\boldsymbol{\xi}_t$ denotes the uncertainty parameter vector for period t, $\boldsymbol{H}_t$ is the coefficient matrix of $\boldsymbol{\xi}_t$ in the uncertainty set constraints (Eqs. (11)~ (16)), $\boldsymbol{z}_t$ represents the constant term vector, $N = J \cdot K + I \cdot K + I \cdot J \cdot K$ equals the number of uncertain parameters per period, and V indicates the number of constraints in the uncertainty set per period.

$$\Xi_t = \{\boldsymbol{\xi}_t = (\boldsymbol{\xi}_t^{\mathrm{Pur}}, \boldsymbol{\xi}_t^{\mathrm{Dem}}, \boldsymbol{\xi}_t^{\mathrm{Tra}}) \in \mathbb{R}^N : \boldsymbol{H}_t \boldsymbol{\xi}_t \le \boldsymbol{z}_t, \boldsymbol{H}_t \in \mathbb{R}^{V \times N}, \boldsymbol{z}_t \in \mathbb{R}^V\} \ \forall t = 1, \cdots, T \tag{17}$$

4.2 Adaptive Feedback Adjustment Rules

After the completion of period t-1, the execution agent reports disclosed uncertainties in resource demand, procurement, and transportation. The decision agent dynamically adjusts the allocation plan for the next period t based on historical uncertainty realizations, thereby adaptively addressing deterministic outcomes from prior periods. As illustrated in **Figure 4**, recourse decision variables in each period t depend on uncertainties $\boldsymbol{\xi}_{[t-\Delta t, t-1]}$ revealed over the preceding Δt periods. Thus, decision variables for resource transportation, procurement, and fund allocation in period t can be expressed as vector functions: $\boldsymbol{y}_t^{\mathrm{Tra}}(\boldsymbol{\xi}_{[t-\Delta t, t-1]}) = (Tra_{t,1,1,1}, \cdots, Tra_{t,I,J,K}) \in \mathbb{R}^{I \cdot J \cdot K}$, $\boldsymbol{y}_t^{\mathrm{Pur}}(\boldsymbol{\xi}_{[t-\Delta t, t-1]}) = (Pur_{t,1,1}, \cdots, Pur_{t,I,K}) \in \mathbb{R}^{I \cdot K}$, and $\boldsymbol{y}_t^{\mathrm{Bud}}(\boldsymbol{\xi}_{[t-\Delta t, t-1]}) = (Bud_{t,1}, \cdots, Bud_{t,I}) \in \mathbb{R}^I$. This study employs affine decision rules [29] to characterize the dependency of recourse decisions on historical uncertainty realizations, as represented by Eqs. (18)~(20). Here, vectors $y_t^{\mathrm{Tra},0}$, $y_t^{\mathrm{Pur},0}$, and $y_t^{\mathrm{Bud},0}$ denote intercept terms representing the initial pre-disaster allocation plan. Matrices $Y_t^{\mathrm{Tra},1}$, $Y_t^{\mathrm{Pur},1}$, and $Y_t^{\mathrm{Pur},1}$ are slope coefficients quantifying adjustment rates in response to disclosed uncertainties.

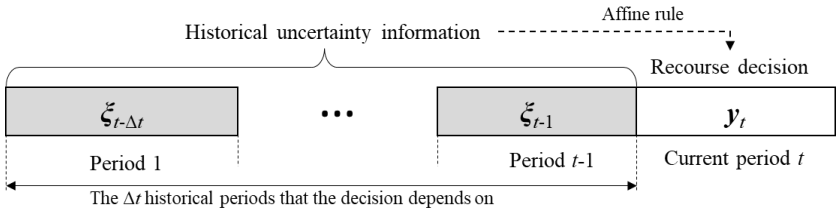

Figure 4 Recourse decision diagram

$$\boldsymbol{y}_t^{\mathrm{Tra}}(\boldsymbol{\xi}_{[t-\Delta t, t-1]}) = \boldsymbol{y}_t^{\mathrm{Tra},0} + \boldsymbol{Y}_t^{\mathrm{Tra},1} \boldsymbol{\xi}_{[t-\Delta t, t-1]} \ \forall t = 2, \cdots, T \tag{18}$$

$$\boldsymbol{y}_t^{\mathrm{Pur}}(\boldsymbol{\xi}_{[t-\Delta t, t-1]}) = \boldsymbol{y}_t^{\mathrm{Pur},0} + \boldsymbol{Y}_t^{\mathrm{Pur},1} \boldsymbol{\xi}_{[t-\Delta t, t-1]} \ \forall t = 2, \cdots, T \tag{19}$$

$$\boldsymbol{y}_t^{\mathrm{Bud}}(\boldsymbol{\xi}_{[t-\Delta t, t-1]}) = \boldsymbol{y}_t^{\mathrm{Bud},0} + \boldsymbol{Y}_t^{\mathrm{Bud},1} \boldsymbol{\xi}_{[t-\Delta t, t-1]} \ \forall t = 2, \cdots, T \tag{20}$$

For conciseness, all recourse decisions in period t are integrated into a single vector, i.e. $\boldsymbol{y}_t(\boldsymbol{\xi}_{[t-\Delta t, t-1]}) \in \mathbb{R}^{P \times 1}$, as shown in Eq. (21). The affine decision rule is then reformulated as Eqs. (22) and (23), where variables $\boldsymbol{y}_t^0 \in \mathbb{R}^{P \times 1}$ and $\boldsymbol{Y}_t^1 \in \mathbb{R}^{P \times (N \cdot \Delta t)}$ denote the compact intercept terms and slope coefficients. The term $\boldsymbol{y}_{t,\tau}^1 \in \mathbb{R}^{P \times N}$ denotes the submatrix of $\boldsymbol{Y}_t^1$ associated with $\boldsymbol{\xi}_\tau$. $P = I \cdot J \cdot K + I \cdot K + I$ refers to the total number of decision variables in period t.

$$\boldsymbol{y}_t(\boldsymbol{\xi}_{[t-\Delta t, t-1]}) = \left(\boldsymbol{y}_t^{\mathrm{Tra}}(\boldsymbol{\xi}_{[t-\Delta t, t-1]}), \boldsymbol{y}_t^{\mathrm{Pur}}(\boldsymbol{\xi}_{[t-\Delta t, t-1]}), \boldsymbol{y}_t^{\mathrm{Bud}}(\boldsymbol{\xi}_{[t-\Delta t, t-1]})\right) \forall t = 2, \cdots, T \tag{21}$$

$$\boldsymbol{y}_t(\boldsymbol{\xi}_{[t-\Delta t, t-1]}) = \boldsymbol{y}_t^0 + \boldsymbol{Y}_t^1 \boldsymbol{\xi}_{[t-\Delta t, t-1]} = \boldsymbol{y}_t^0 + \sum_{\tau=\max(1, t-\Delta t)}^{t-1} \boldsymbol{y}_{t,\tau}^1 \boldsymbol{\xi}_\tau \ \forall t = 2, \cdots, T \tag{22}$$

$$\boldsymbol{y}_t^0 = (\boldsymbol{y}_t^{\mathrm{Tra},0}, \boldsymbol{y}_t^{\mathrm{Pur},0}, \boldsymbol{y}_t^{\mathrm{Bud},0}), \boldsymbol{Y}_t^1 = (\boldsymbol{Y}_t^{\mathrm{Tra},1}, \boldsymbol{Y}_t^{\mathrm{Pur},1}, \boldsymbol{Y}_t^{\mathrm{Bud},1}) \tag{23}$$

4.3 Compact Multi-Stage Robust Optimization Model

Building upon the deterministic resource allocation model (Section 3), periodic uncertainty sets (Section 4.1), and feedback-based decision rules (Section 4.2), we establish an integrated multi-timescale, multi-agent collaborative resource allocation model. To facilitate subsequent linearization, we present the model using matrix-vector notation. Firstly, the original comprehensive objective (Eq. (10)) is augmented with recourse decision variables, yielding Eq. (24). Here, $\boldsymbol{x} = (a_{1,1,1}, \cdots, a_{t,j,k}, \cdots, a_{T,J,K})^{\mathrm{T}} \in \mathbb{R}^{F \times 1}$ $(F = T \cdot J \cdot K)$ is the auxiliary variable vector for supply-demand gaps and $\boldsymbol{c} = (1, \cdots, 1)^{\mathrm{T}} \in \mathbb{R}^{F \times 1}$ is its coefficient vector. $\boldsymbol{q}_t \in \mathbb{R}^{P \times 1}$ is the coefficient vector for recourse decisions $\boldsymbol{y}_t(\boldsymbol{\xi}_{[\tau-\Delta t, t-1]})$. Ξ_t denotes the uncertainty set for period t (Eq. (17)).

$$\min_{\boldsymbol{x}, \boldsymbol{y}_t (1 \le t \le T)} \left\{\boldsymbol{c}^{\mathrm{T}} \boldsymbol{x} + \boldsymbol{q}_1^{\mathrm{T}} \boldsymbol{y}_1 + \max_{\boldsymbol{\xi}_t \in \Xi_t (1 \le t \le T-1)} \sum_{t=2}^{T} \boldsymbol{q}_t^{\mathrm{T}} \boldsymbol{y}_t(\boldsymbol{\xi}_{[t-\Delta t, t-1]})\right\} \tag{24}$$

Secondly, incorporating uncertainty parameters $\boldsymbol{\xi}_t \in \Xi_t$ and recourse decisions $\boldsymbol{y}_t(\boldsymbol{\xi}_{[\tau-\Delta t, t-1]})$ transforms the original constraints (Eqs. (1)~(4)~(9)) into Eqs. (25) and (26). Here, Eq. (25) involves Period 1 Constraints without historical uncertainty and dynamic adjustments. Eq. (26) denotes Period t≥2 Constraints with historical uncertainty and corresponding dynamic adjustments. In addition, $\boldsymbol{\xi}_{[t]} = (\boldsymbol{\xi}_1, \cdots, \boldsymbol{\xi}_t)$ aggregates uncertainties up to period t. $\boldsymbol{y}_t(\boldsymbol{\xi}_{[\tau-\Delta t, t-1]})$ follows Eqs. (22) and (23). $\boldsymbol{W}_{t,\tau} \in \mathbb{R}^{M \times P}, \boldsymbol{T}_t \in \mathbb{R}^{M \times F}, \boldsymbol{B}_t \in \mathbb{R}^{M \times (N \cdot t)}$ are separately coefficient matrices regarding $\boldsymbol{y}_t, \boldsymbol{x}, \boldsymbol{\xi}_{[t]}$ in original constraints. $\boldsymbol{u}_t \in \mathbb{R}^{M \times 1}$ is the

corresponding constant vector. M denotes the number of constraints per period.

$$\boldsymbol{W}_{1,1}\boldsymbol{y}_1 + \boldsymbol{T}_1\boldsymbol{x} \geq \boldsymbol{B}_1\boldsymbol{\xi}_1 + \boldsymbol{u}_1 \ \forall \boldsymbol{y}_1 \in \mathbb{R}^{P\times 1}, \boldsymbol{\xi}_1 \in \Xi_1 \tag{25}$$

$$\boldsymbol{W}_{1,1}\boldsymbol{y}_1 + \sum_{\tau=2}^{t}\boldsymbol{W}_{t,\tau}\boldsymbol{y}_\tau\left(\boldsymbol{\xi}_{[\tau-\Delta t,\tau-1]}\right) + \boldsymbol{T}_t\boldsymbol{x} \geq \boldsymbol{B}_t\boldsymbol{\xi}_{[t]} + \boldsymbol{u}_t \ \forall \boldsymbol{y}_t\left(\boldsymbol{\xi}_{[\tau-\Delta t,t-1]}\right) \in \mathbb{R}^{P\times 1}, \boldsymbol{\xi}_t \in \Xi_t, t = 2,\cdots,T \tag{26}$$

5 Linear Reformulation of the Robust Allocation Model

The robust optimization model for dynamic resource allocation contains uncertain parameters and semi-infinite constraints, rendering it computationally intractable. To address this challenge, we apply linear programming duality principles and semi-infinite constraint techniques [30], transforming the original MSARO model into a tractable linear programming formulation. This reformulation reduces computational complexity while preserving the model's core properties, ensuring solution accuracy and robustness.

(1) Step 1: Incorporate Recourse Decision Expressions

Substituting the affine decision rules (Eqs. (22) and (23)) into the objective function (Eq. (24)) yields Eq. (27). Similarly, substituting into constraints (Eqs. (25) and (26)) and rearranging terms gives Eq. (28), where $\boldsymbol{B}'_{t,\tau} \in \mathbb{R}^{M\times N}$ is the submatrix of $\boldsymbol{B}_t$ associated with $\boldsymbol{\xi}_\tau$.

$$\min_{\substack{\boldsymbol{x},\boldsymbol{y}_1,\boldsymbol{y}_t^0,\boldsymbol{y}_{t,\tau}^1 \\ (1\leq t\leq T)}} \left\{\boldsymbol{c}^{\mathrm{T}}\boldsymbol{x} + \boldsymbol{q}_1^{\mathrm{T}}\boldsymbol{y}_1 + \sum_{t=2}^{T}\boldsymbol{q}_t^{\mathrm{T}}\boldsymbol{y}_t^0 + \max_{\substack{\boldsymbol{\xi}_t\in\Xi_t \\ (1\leq t\leq T-1)}} \sum_{t=2}^{T}\boldsymbol{q}_t^{\mathrm{T}}\sum_{\tau=\max(1,t-\Delta t)}^{t-1}\boldsymbol{y}_{t,\tau}^1\boldsymbol{\xi}_\tau\right\} \tag{27}$$

$$\begin{cases} \boldsymbol{W}_{1,1}\boldsymbol{y}_1 + \boldsymbol{T}_1\boldsymbol{x} \geq \boldsymbol{B}_1\boldsymbol{\xi}_1 + \boldsymbol{u}_1 \ \forall \boldsymbol{\xi}_1 \in \Xi_1 \ \ (a) \\ \boldsymbol{W}_{1,1}\boldsymbol{y}_1 + \sum_{\tau=2}^{t}\boldsymbol{W}_{t,\tau}\boldsymbol{y}_\tau^0 + \boldsymbol{T}_t\boldsymbol{x} - \boldsymbol{u}_t \geq \sum_{\tau=1}^{t-1}\boldsymbol{B}'_{t,\tau}\boldsymbol{\xi}_\tau - \sum_{\tau=2}^{t}\boldsymbol{W}_{t,\tau}\sum_{\tau'=\max(1,\tau-\Delta t)}^{\tau-1}\boldsymbol{y}_{\tau,\tau'}^1\boldsymbol{\xi}_{\tau'} + \boldsymbol{B}'_{t,t}\boldsymbol{\xi}_t \\ \forall \boldsymbol{\xi}_t \in \Xi_t, 2\leq t\leq T \ \ (b) \end{cases} \tag{28}$$

(2) Step 2: Embed Uncertainty Set Expressions

Integrating the uncertainty set Ξ_t (Eq. (17)) into Eq. (27) gives Eq. (29). Similarly, the semi-infinite constraints in Eq. (28) are equivalently reformulated as the maximization problem in the right side of Eqs. (30) and (31) [31].

$$\min_{\substack{\boldsymbol{x},\boldsymbol{y}_1,\boldsymbol{y}_t^0,\boldsymbol{y}_{t,\tau}^1 \\ (1\leq t\leq T)}} \left\{\boldsymbol{c}^{\mathrm{T}}\boldsymbol{x} + \boldsymbol{q}_1^{\mathrm{T}}\boldsymbol{y}_1 + \sum_{t=2}^{T}\boldsymbol{q}_t^{\mathrm{T}}\boldsymbol{y}_t^0 + \max_{\substack{\boldsymbol{\xi}_t\in\mathbb{R}^N \\ (1\leq t\leq T-1)}} \left\{\sum_{t=1}^{T-1}\sum_{\tau=t+1}^{\min(t+\Delta t,T)}\boldsymbol{q}_\tau^{\mathrm{T}}\boldsymbol{y}_{\tau,t}^1\boldsymbol{\xi}_t : \boldsymbol{H}_t\boldsymbol{\xi}_t \leq \boldsymbol{z}_t, 1\leq t \leq T-1\right\}\right\} \tag{29}$$

$$\boldsymbol{W}_{1,1}\boldsymbol{y}_1 + \boldsymbol{T}_1\boldsymbol{x} - \boldsymbol{u}_1 \geq \max_{\boldsymbol{\xi}_1\in\mathbb{R}^N}\{\boldsymbol{B}_1\boldsymbol{\xi}_1 : \boldsymbol{H}_1\boldsymbol{\xi}_1 \leq \boldsymbol{z}_1\} \tag{30}$$

$$\boldsymbol{W}_{1,1}\boldsymbol{y}_1 + \sum_{\tau=2}^{t}\boldsymbol{W}_{t,\tau}\boldsymbol{y}_\tau^0 + \boldsymbol{T}_t\boldsymbol{x} - \boldsymbol{u}_t \geq \max_{\substack{\boldsymbol{\xi}_\tau\in\mathbb{R}^N \\ (1\leq\tau\leq t)}} \left\{\sum_{\tau=1}^{t-1}(\boldsymbol{B}'_{t,\tau} - \sum_{\tau'=\tau+1}^{\min(t,\tau+\Delta t)}\boldsymbol{W}_{t,\tau'}\boldsymbol{y}_{\tau',\tau}^1)\boldsymbol{\xi}_\tau + \boldsymbol{B}'_{t,t}\boldsymbol{\xi}_t : \boldsymbol{H}_\tau\boldsymbol{\xi}_\tau \leq \boldsymbol{z}_\tau, 1\leq\tau\leq t\right\} 2\leq t\leq T \tag{31}$$

(3) Step 3: Robust Dual Transformation

Applying strong duality theory [32], we introduce dual variables $\boldsymbol{\beta}_t \in \mathbb{R}_+^V$, $\boldsymbol{\alpha}_1 \in \mathbb{R}_+^{V\times M}$ and $\boldsymbol{\alpha}_{t,\tau} \in \mathbb{R}_+^{V\times M}$ for Eqs. (29)~(31), respectively. The max terms in Eqs. (29)~(31) are equivalently transformed into minimization problems, namely Eqs. (32)~(34).

$$\min_{\substack{\boldsymbol{x},\boldsymbol{y}_1,\boldsymbol{y}_t^0,\boldsymbol{y}_{t,\tau}^1 \\ (1\leq t\leq T)}} \left\{\boldsymbol{c}^{\mathrm{T}}\boldsymbol{x} + \boldsymbol{q}_1^{\mathrm{T}}\boldsymbol{y}_1 + \sum_{t=2}^{T}\boldsymbol{q}_t^{\mathrm{T}}\boldsymbol{y}_t^0 + \min_{\substack{\boldsymbol{\beta}_t\in\mathbb{R}_+^V \\ (1\leq t\leq T-1)}} \left\{\sum_{t=1}^{T-1}\boldsymbol{z}_\tau^{\mathrm{T}}\boldsymbol{\beta}_t : \boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{\beta}_t = \sum_{\tau=t+1}^{\min(T,t+\Delta t)}\boldsymbol{y}_{\tau,t}^{1\ \mathrm{T}}\boldsymbol{q}_\tau, 1\leq t \leq T-1\right\}\right\} \tag{32}$$

$$\boldsymbol{W}_{1,1}\boldsymbol{y}_1 + \boldsymbol{T}_1\boldsymbol{x} - \boldsymbol{u}_1 \geq \min_{\boldsymbol{\alpha}_1\in\mathbb{R}_+^{V\times M}}\{\boldsymbol{\alpha}_1^T\boldsymbol{z}_1 : \boldsymbol{H}_1^T\boldsymbol{\alpha}_1 = \boldsymbol{B}_1^T\} \tag{33}$$

$$\boldsymbol{W}_{1,1}\boldsymbol{y}_1 + \sum_{\tau=2}^{t}\boldsymbol{W}_{t,\tau}\boldsymbol{y}_\tau^0 + \boldsymbol{T}_t\boldsymbol{x} - \boldsymbol{u}_t \geq \min_{\substack{\boldsymbol{\alpha}_{t,\tau}\in\mathbb{R}_+^{V\times M} \\ (1\leq\tau\leq t)}} \left\{\sum_{\tau=1}^{t}\boldsymbol{\alpha}_{t,\tau}^{\mathrm{T}}\boldsymbol{z}_\tau : \begin{cases} \boldsymbol{H}_\tau^{\mathrm{T}}\boldsymbol{\alpha}_{t,\tau} = \boldsymbol{B}'^{\mathrm{T}}_{t,\tau} - \sum_{\tau'=\tau+1}^{\min(t,\tau+\Delta t)}\left(\boldsymbol{W}_{t,\tau'}\boldsymbol{y}_{\tau',\tau}^1\right)^{\mathrm{T}}, 1\leq\tau\leq t-1 \\ \boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{\alpha}_{t,t} = \boldsymbol{B}'^{\mathrm{T}}_{t,t} \end{cases}\right\} 2\leq t\leq T \tag{34}$$

(4) Step 4: Final Linear Programming Form

Splitting the nested minimization in Eq. (32) gives the decoupled objective and constraint, i.e. Eqs. (35) and (36). Similarly, Eqs. (33) and (34) simplify to Eqs. (37)~(41). Up to now, the original MSARO model is

now reformulated as the tractable linear program (Eqs. (35)~(41)) with variables: $\boldsymbol{x} \in \mathbb{R}^{F\times 1}, \boldsymbol{y}_1 \in \mathbb{R}^{P\times 1}, \boldsymbol{y}_t^0 \in \mathbb{R}^{P\times 1}, \boldsymbol{y}_{t,\tau}^1 \in \mathbb{R}^{P\times N} (\tau = \max(1, t-\Delta t), \cdots, t-1; t = 2, \cdots, T), \boldsymbol{\beta}_t \in \mathbb{R}_+^V, \boldsymbol{\alpha}_1 \in \mathbb{R}_+^{V\times M}, \boldsymbol{\alpha}_{t,\tau} \in \mathbb{R}_+^{V\times M} (\tau = 1, \cdots, t-1; t = 2, \cdots, T)$.

$$\min_{\boldsymbol{x}, \boldsymbol{y}_1, \boldsymbol{y}_t^0, \boldsymbol{\beta}_t} \{\boldsymbol{c}^{\mathrm{T}}\boldsymbol{x} + \boldsymbol{q}_1^{\mathrm{T}}\boldsymbol{y}_1 + \sum_{t=2}^{T} \boldsymbol{q}_t^{\mathrm{T}}\boldsymbol{y}_t^0 + \sum_{t=1}^{T-1} \boldsymbol{z}_\tau^{\mathrm{T}}\boldsymbol{\beta}_t\} \quad (1 \le t \le T) \tag{35}$$

$$\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{\beta}_t = \sum_{\tau=t+1}^{\min(T,t+\Delta t)} \boldsymbol{y}_{\tau,t}^{1\ \mathrm{T}}\boldsymbol{q}_\tau \ \forall t = 1, \cdots, T-1 \tag{36}$$

$$\boldsymbol{W}_{1,1}\boldsymbol{y}_1 + \boldsymbol{T}_1\boldsymbol{x} - \boldsymbol{u}_1 \ge \boldsymbol{\alpha}_1^T\boldsymbol{z}_1 \tag{37}$$

$$\boldsymbol{H}_1^T\boldsymbol{\alpha}_1 = \boldsymbol{B}_1^T \tag{38}$$

$$\boldsymbol{W}_{1,1}\boldsymbol{y}_1 + \sum_{\tau=2}^{t} \boldsymbol{W}_{t,\tau}\boldsymbol{y}_\tau^0 + \boldsymbol{T}_t\boldsymbol{x} - \boldsymbol{u}_t \ge \sum_{\tau=1}^{t} \boldsymbol{\alpha}_{t,\tau}^{\mathrm{T}}\boldsymbol{z}_\tau \ \forall t = 2, \cdots, T \tag{39}$$

$$\boldsymbol{H}_\tau^{\mathrm{T}}\boldsymbol{\alpha}_{t,\tau} = \boldsymbol{B}'^{\mathrm{T}}_{t,\tau} - \sum_{\tau'=\tau+1}^{\min(t,\tau+\Delta t)} \left(\boldsymbol{W}_{t,\tau'}\boldsymbol{y}_{\tau',\tau}^1\right)^{\mathrm{T}} \ \forall \tau = 1, \cdots, t-1; t = 2, \cdots, T \tag{40}$$

$$\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{\alpha}_{t,t} = \boldsymbol{B}'^{\mathrm{T}}_{t,t} \ \forall t = 2, \cdots, T \tag{41}$$

6. Experiment Analysis and Discussion

6.1 Case Study Description

Earthquakes represent frequent natural disasters that severely threaten human lives and property. This case study examines a major earthquake event causing surface ruptures, landslides, and debris flows that critically damaged transportation infrastructure. Post-disaster, local authorities established an emergency command center as the decision gent to coordinate response efforts, including fund allocation. Two municipal transfer stations (TS#1–2) serves as the execution agent. The demand agent involves two temporary shelters in severely disaster areas (DA#1–2). **Figure 5** illustrates the geographical distribution and resource flow directions. **Table 1** details the affected populations and transportation distances [33].

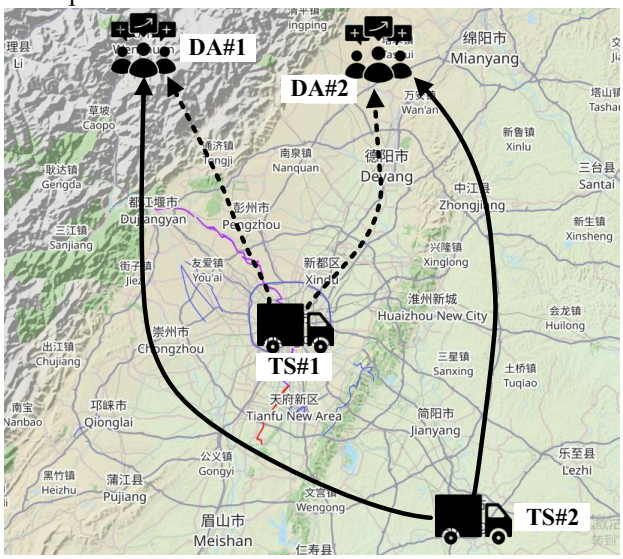

Figure 5 Schematic diagram of post-earthquake emergency resource allocation from transfer stations to disaster areas

Table 1 Affected populations and transportation distances

Transfer station /Disaster area	DA#1	DA#2
TS#1（km）	143	211
TS#2（km）	177	177
Affected population	34583	26916

(1) Parameter Configuration

Standard 2-ton trucks are used for resource transportation. Referencing highway toll standards [34] and accounting for road damage and repair progress, transportation costs per vehicle are set as shown in **Table 2**. Roads suffer maximum damage on Day 1, resulting in the highest transportation costs. Costs gradually decrease as repairs progress. The transportation loss compensation ratio Rat is set to 0.05.

Table 2 Post-disaster highway tolls

Post-disaster time	Day 1	Day 2	Day 3
Toll per truck (¥/km)	0.8	0.7	0.6

Earthquake relief resources include first-aid kits, instant food, and drinking water [35] to ensure life-saving utility. First-aid kits are single-use resources (one kit satisfies one survivor's needs during the golden rescue window), while food and water are daily consumables (each survivor requires daily rations). Resource

specifications are provided in **Table 3**.

Table 3 Emergency resource standards

Resource type	Resource	Cost	Loading capacity	Allocation standard
1	First-aid kits	¥160/unit	600 units	1 unit/person
2	Instant food	¥5/bag	4,000 bags	6 bags/person/day
3	Drinking water	¥3/L	2,000 L	2 L/person/day

To eliminate decision bias from subjective preferences, marginal life-saving utility (MLSU) is calculated using the entropy method [36]. Disaster areas farther from transfer stations yield higher MLSU per unit resource. On Day 1, with limited disaster information, MLSU is estimated based on average distances. For subsequent days, parameters are updated using the same method after collecting site-specific data. MLSU values during the golden rescue window are listed in **Table 4** [27].

Table 4 Marginal life-saving utility per disaster area

Disaster areas	Day 1	Day 2	Day 3
DA#1	0.4	1.0	2.82
DA#2	0.5	2.28	1.21

Without loss of generality, resource demand, procurement, and transportation uncertainties follow normal distributions, i.e. $\xi_{t,j,k}^{\mathrm{Dem}} \sim \mathcal{N}(0, \sigma_{t,j,k}^{\mathrm{Dem}^2})$, $\xi_{t,i,k}^{\mathrm{Pur}} \sim \mathcal{N}(0, \sigma_{t,i,k}^{\mathrm{Pur}^2})$ and $\xi_{t,i,j,k}^{\mathrm{Tra}} \sim \mathcal{N}(0, \sigma_{t,i,j,k}^{\mathrm{Tra}\ 2})$. The ratios of uncertainty parameter bounds to predicted values are set as shown in **Table 5**, which makes uncertainty sets cover 95% of the uncertainty distribution. Referring to the literature [37], initial information scarcity causes maximal uncertainty on Day 1. Uncertainty significantly decreases over subsequent days as damage assessments improve and infrastructure recovers. Rescue operations progressively confirm survivor counts, increasing daily resource demands. Note that the uncertain deviations of procurement and transportation are restricted to non-positive values.

Table 5 Uncertainty bounds (ratio to estimates)

Post-disaster time	Demand	Procurement	Transportation
Day 1	[0.6, 0.8]	[0.8, 1.0]	[0.8, 1.0]
Day 2	[0.8, 0.95]	[0.9, 1.0]	[0.9, 1.0]
Day 3	[0.95, 1.05]	[0.95, 1.0]	[0.95, 1.0]

(2) Model Execution and Evaluation Metrics

Parameters are set as Δt=1and T=3, discretizing the 72-hour golden rescue window into three days. Dynamic allocation strategies are implemented based on prior-day uncertainty feedback. Monte Carlo simulation generated 1,000 uncertainty samples for testing. For each sample, model-derived allocation strategies were executed to validate stability and adaptability. The model was implemented in Python 3.8 using Gurobipy (v11.2) with a runtime limit of 5 minutes. All tests ran on Windows 11 with a 6-core CPU (AMD Ryzen 5 5600G, 3.90 GHz) and 32 GB RAM.

Several key metrics are adopted to evaluate comprehensive model performance: 1) Comprehensive optimization objective: Mean and standard deviation of the comprehensive objective across all samples reflect the optimality and stability; 2) Model scale: Variable and constraint counts have an impact on model computation efficiency; 3) Solution time: Evaluate model computation efficiency and directly impact the timeliness of post-disaster response; 4) Constraint violation rate: Proportion of samples that violate constraints measures the safety and reliability of the allocation decision.

6.2 Analysis of Emergency Resource Allocation Results

Based on the case setup above, the allocation process and optimization results of the model under uncertainty disturbances are presented. The analysis focuses on resource demand response, multi-timescale management of funds and resources, multi-agent optimization objectives, and validates the model's dynamic adaptability, optimization effectiveness, and robustness in handling uncertainties.

(1) Dynamics and Uncertainty of Emergency Resource Demand

Figure 6 clearly illustrates the predicted demand values and the actual distribution under uncertainty samples for three types of emergency resources at the two disaster areas (DA#1 and DA#2) during the 72-hour post-disaster period. On Day 1, the uncertainty range for resource demand is the largest (approximately ±20%), consistent with the initial severe lack of disaster information (see **Table 5**). As disaster information becomes progressively clearer on Days 2 and 3 (through updated loss assessments and population statistics), the uncertainty range narrows significantly (within ±5%), validating the parameter settings. Uncertainty in demand for single-use resources (first-aid kits) is primarily concentrated on Day 1. In contrast, uncertainty for daily-consumption resources (instant food and drinking water) persists throughout the golden rescue period and correlates with population estimation errors. This occurs because only newly disclosed affected populations will receive supplementary first-aid kits on subsequent days, while daily resource consumption increases as the affected population is progressively revealed. DA#1 (with a larger population) generally exhibits higher resource demand quantities and uncertain fluctuation ranges than DA#2. This significant time-varying

uncertainty and resource demand disparity provide robust testing conditions for evaluating model performance.

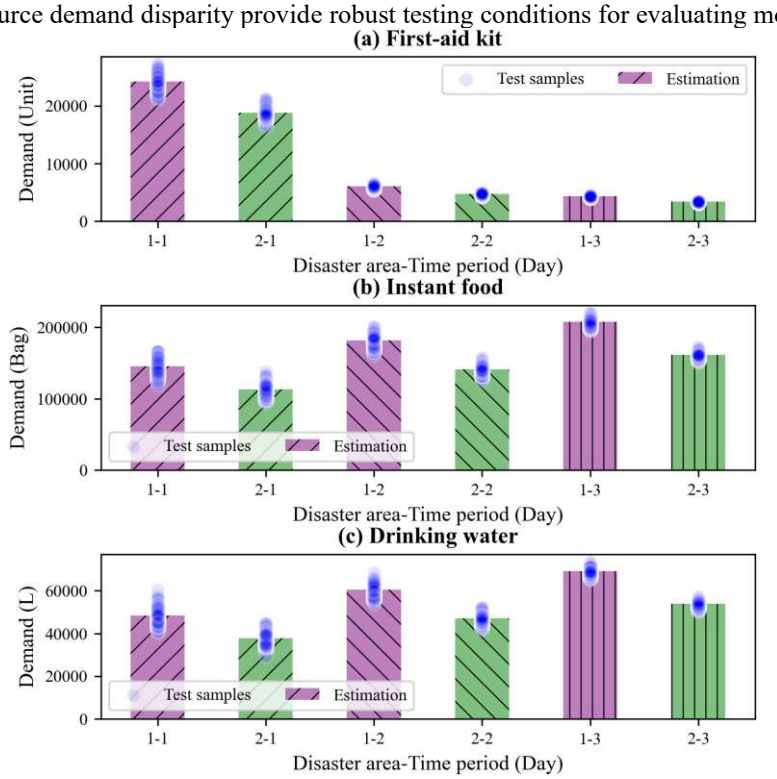

Figure 6 Resource demand at disaster areas (pre-disaster estimates vs. actual sample values)

(2) Analysis of Multi-Agent and Multi-Timescale Collaborative Allocation

Figure 7~**Figure 9** demonstrate that the model not only formulates a global allocation plan in the pre-disaster phase (through optimization solution) but also achieves dynamic coordinated allocation across the "Decision-Execution-Demand" triad after the disaster occurs (via recourse decisions based on daily uncertainty updates in each scenario). **Figure 7** shows the periodic and differentiated fund allocation strategy of the decision agent. Fund allocation to the two transfer stations (TS#1-2) is highest on Day 1 with no dynamic adjustment, addressing the initial high uncertainty and urgent procurement/transportation demands. Subsequent allocations are dynamically adjusted based on the previous day's feedback. For instance, if sample results on Day 1 indicate higher-than-expected actual demand at DA#1 or increase transportation shortfalls from TS#1 to DA#1, the allocation to TS#1 on Day 2 will exceed the pre-planned budget, with minor adjustments to TS#2. Fund flow, as the starting point for the collaborative allocation, establishes the foundation for subsequent stages through its dynamic adjustments.

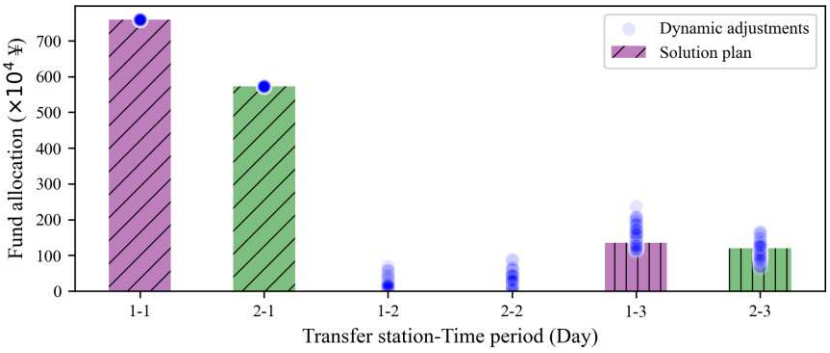

Figure 7 Fund allocation by the decision agent (pre-disaster plan vs. dynamic adjustments in post-disaster samples)

Figure 8 illustrates that resource procurement by the execution agent is closely coordinated with fund allocation and dynamically updated based on resource demand. On Day 1, TS#1-2 procures resources according to the initial demand estimates to meet urgent needs. Subsequent procurement adjustments are rolled forward based on fund availability and actual demand/procurement quantities from prior periods. For example, if procurement difficulties at TS#1 occurs on Day 2, TS#1 moderately increases procurement of that resource on Day 3, or TS#2 shares the procurement task. The procurement strategy effectively links the funds allocated by the decision agent with resource transportation by the execution agent.

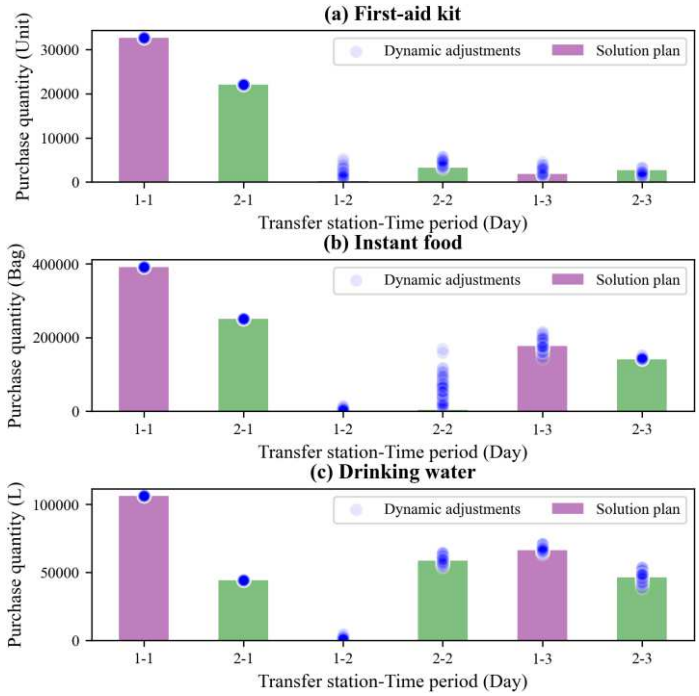

Figure 8 Resource procurement at transfer stations by the execution agent (pre-disaster plan vs. dynamic adjustments in post-disaster samples)

Resource transportation directly links available resources (procurement minus inventory) to dynamic demand. **Figure 9** demonstrates the execution agent's tendency to prioritize routes with higher MLSU and the dynamic adjustment of transportation quantities. For instance, if increased transportation shortfalls occur on the TS#1→DA#1 route, the shipment quantity on this route will be reduced in subsequent periods compared to the plan, with some capacity shifted to TS#1→DA#2 or TS#2→DA#1 route. More importantly, transportation quantities are promptly corrected based on the actual demand reported by the demand agent. For example, if actual resource consumption at DA#1 on Day 1 exceeds expectations, the execution agent will immediately increase resource shipments to DA#1 on Day 2. This demand- and procurement-feedback-based dynamic transportation is key to enhancing rescue utility and reducing supply-demand imbalance, fully demonstrating the synergy between execution and demand agents and the efficacy of dynamic allocation.

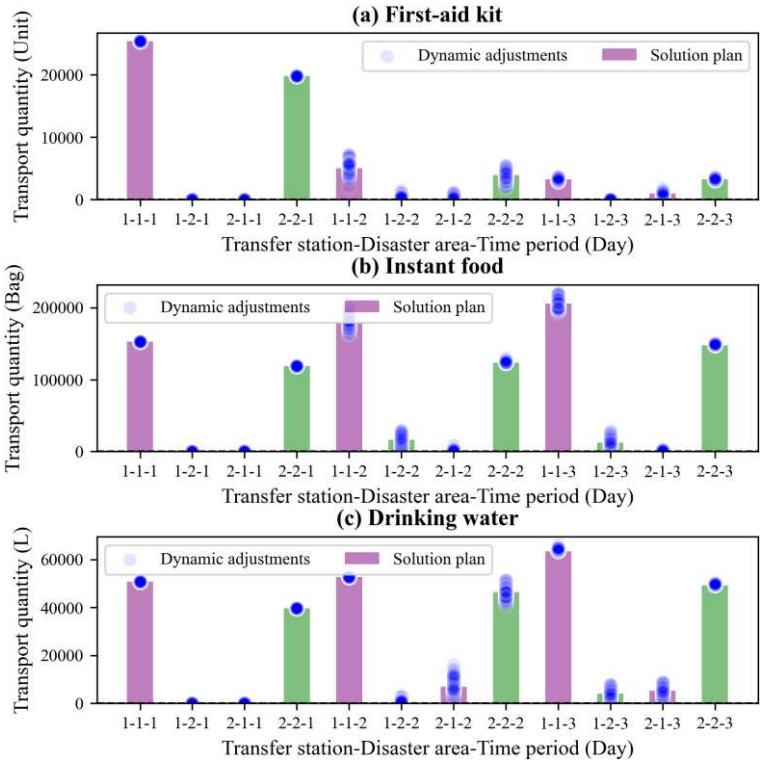

Figure 9 Resource transportation from execution to demand agents (pre-disaster plan vs. dynamic adjustments in post-disaster samples)

In summary, the collaborative allocation keeps the initial plan on Day 1, and the feedback adjustment is implemented from Day 2. The adjustment direction (increase/decrease) and magnitude are highly dependent on the disclosed uncertainty information, demonstrating the effectiveness of the post-disaster feedback adjustment rule. Note that procurement and transportation of first-aid kits peak on Day 1, while those of food and water increase daily, as illustrated in **Figure 8** and **Figure 9**. This occurs because, for the next two days, first-aid kits are supplied in small quantities as supplements to newly disclosed populations, whereas food and water are required daily for all growing affected populations.

(3) Dynamic Evolution of Key States and Robustness

The trajectories of cumulative fund balance, resource inventory surplus, and supply-demand gap are key state variables that validate the robustness of the model allocation. According to **Figure 10**, the cumulative fund balance at each transfer station remains non-negative across most test samples, almost satisfying the core constraint (Eq. (1)). The planned daily balances for TS#1 and TS#2 approaches zero, striving for supply-demand equilibrium in fund flow to control economic costs. Although uncertainty causes fluctuations in the actual balance, effectively adjusted fund allocation and resource procurement strategies maintain fund chain security, preventing allocation disruption due to fund depletion and ensuring the economic sustainability of resource allocation.

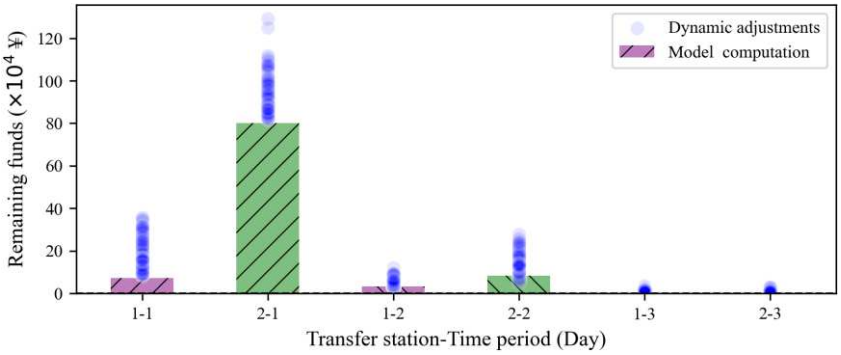

Figure 10 Dynamic fund balance at transfer stations (pre-disaster plan vs. dynamic adjustments in post-disaster samples)

Figure 11 shows that the inventory surplus for each resource type at each transfer station remains non-negative across most samples, almost satisfying constraint (Eq. (2)). Inventory dynamics exhibit reasonable fluctuations: higher surpluses on Day 1 to establish safety stock against uncertainty; as demand information clarifies and procurement/transportation stabilizes, inventory gradually decreases in later periods but is almost maintained at safe levels, indicating the model's buffering capacity and procurement-transportation synergy.

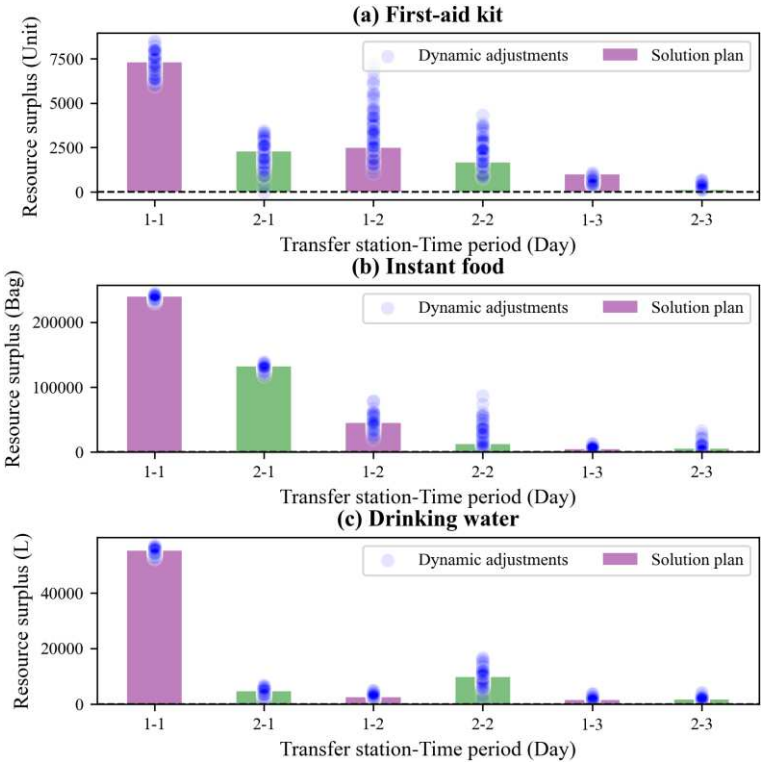

Figure 11 Dynamic emergency resource surplus at transfer stations (pre-disaster plan vs. dynamic adjustments in post-disaster samples)

According to **Figure 12**, although the original plan aims for resource surplus at disaster areas, significant resource shortages or local surpluses occur at both DA#1 and DA#2 in test samples during the initial phase due to high information uncertainty. With information feedback and dynamic adjustment of resource transportation, the resource surplus at each disaster area converges towards the ideal equilibrium point (zero) from Day 2 onwards, with fluctuation ranges significantly narrowing. This indicates a substantial improvement in supply-demand matching, mitigating resource shortages and waste at disaster areas.

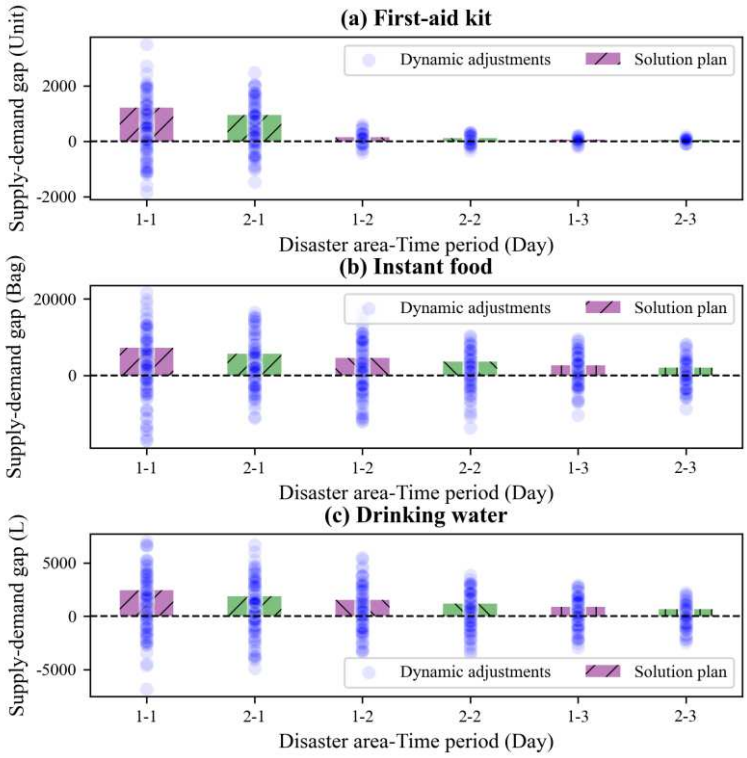

Figure 12 Dynamic resource supply-demand balance at disaster areas (pre-disaster plan vs. dynamic adjustments in post-disaster samples)

(4) Evaluation of Comprehensive Optimization Effectiveness

As shown in **Figure 13**, the comprehensive objective values across all test samples are significantly better than the original calculated value under the allocation model. The tested economic costs are higher than the model computation. That's primarily because the adjustments by top-level fund allocation are relatively sensitive to the uncertainties in disaster areas. The tested life-saving utilities and resource supply-demand gaps exhibit stronger optimality than the model computation, benefiting from transportation strategies dynamically adjusted based on updated actual demand/transportation. Consequently, the allocation model demonstrates favorable optimality on the comprehensive objective.

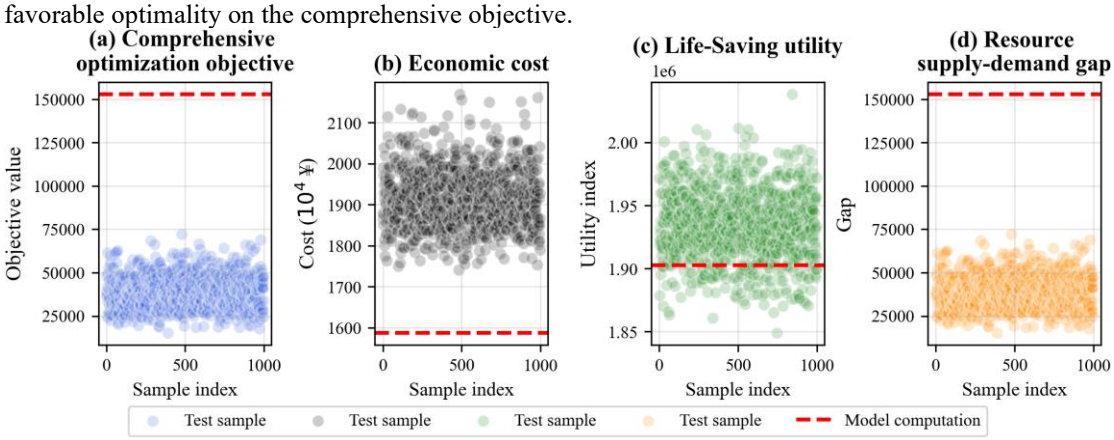

Figure 13 Optimization objectives (model calculation vs. test samples)

6.3 Sensitivity Analysis of Key Parameters

This section investigates the impact of two core parameters, the number of time periods (T) and the decision dependency depth (Δt), on the model's overall performance.

(1) Impact Analysis of the Number of Time Periods (T)

Based on the case setup in Section 6.1 (Δt=1), we analyze how model performance metrics change under different numbers of time periods (T=3, 4, 6, 9, and 12). As shown in **Figure 14** (a), the total number of variables and constraints in the model increases with T, signifying growth in model scale and complexity. This arises because each additional time period introduces a number of decision variables and associated constraints. **Figure 14** (b) shows that the model solution time increases with T. This decrease in solving efficiency stems from the escalating model scale and complexity. Models with ($T\leqslant 9$) are solved effectively. However, for (T=12), the excessive model scale and high computational complexity prevent convergence to the optimal solution within the preset time limit (marked as "Timeout"). Finer time granularity (larger T) captures more detailed disaster dynamics but at the cost of drastically increased computational complexity. Therefore, practical deployment requires balancing response speed with available computational resources. **Figure 14** (c) shows that the sample mean and standard deviation of the comprehensive objective f decrease significantly within the (T=3 to 6) range, indicating that more frequent decision adjustments substantially improved overall scheduling effectiveness. However, as T increases further to (9 and 12), the improvement in the mean f diminishes sharply, while its standard deviation stabilizes. For T=12, they even slightly increase due to the non-optimal solution. According to **Figure 14** (d), the proportion of samples violating core constraints (Eqs. (1) and (2)) are highest at T=3. As T increases, the violation rate gradually decreases, albeit at a diminishing rate. Increasing the frequency of decision adjustments allows for more timely responses to uncertainty shocks. This effectively mitigates or compensates for risks like resource shortages or fund depletion, thereby enhancing solution feasibility and robustness.

Consequently, a significant benefit-cost trade-off exists for the number of periods T. T=6 (i.e., one decision period every 12 hours) represents the "performance inflection point" for this case study: it achieves significant gains in optimization and stability while maintaining efficient solving (within 10 seconds) and reducing violation risk to a very low level. Finer partitioning (T> 6) yields diminishing benefits, offset by drastically increased computational costs and potential solution failure, making it potentially counterproductive.

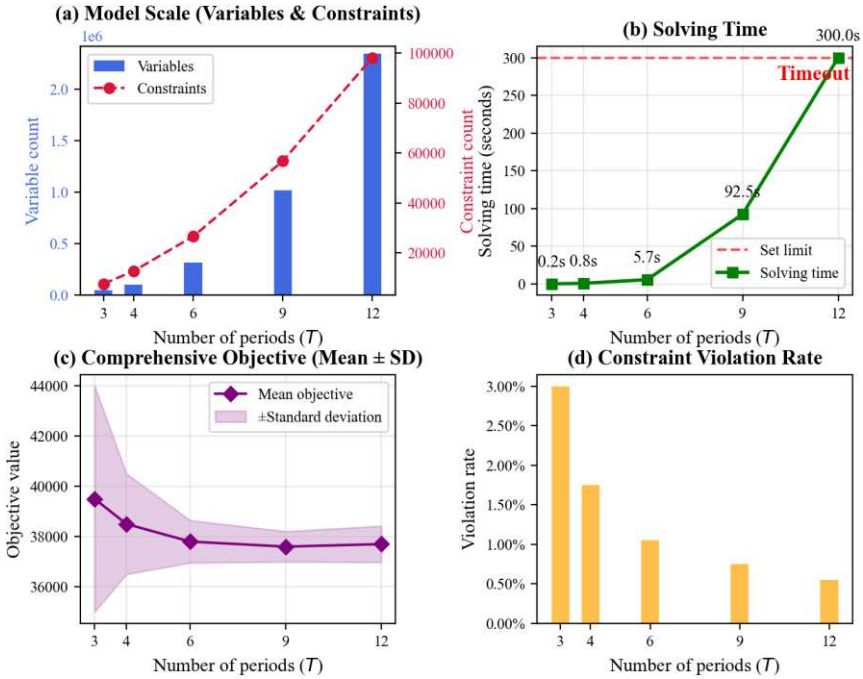

Figure 14 Comprehensive performance metrics under different numbers of time periods (T)

(2) Impact Analysis of Decision Dependency Depth (Δt)

The parameter Δt defines the depth of historical uncertainty information used in the dynamic adjustment rules (Eq. (22)). Based on the previous case setup with T=6, we set Δt to 1, 2, 3, 4, and 5, analyzing changes in model performance metrics. As shown in **Figure 15** (a), the number of model variables increases with Δt, while the constraint count remains unchanged. That's because increasing Δt will introduce more affine slope variables and non-zero terms in original constraints. **Figure 15** (b) shows that the solution time increases with Δt, but this growth is significantly milder compared to the impact of increasing T. Increasing the historical dependency depth Δt enlarges the model scale and extends solving time, but the pressure on computational resources is far less than that caused by increasing the number of time periods T. **Figure 15** (c) shows that the sample mean and standard deviation of the comprehensive objective f continuously decrease as Δt increase, albeit with diminishing returns. This indicates that incorporating more historical information enhances decision stability and optimization quality. This improvement occurs because a longer historical window helps the model capture uncertainty propagation patterns (e.g., persistent demand deviations and route impacts). However, the benefit of excessive historical dependency diminishes sharply, as distant information becomes less relevant or introduces noise. According to **Figure 15** (d), the sample violation rate decreases continuously with increasing Δt. By leveraging more historical uncertainty information, the model can more accurately anticipate potential risks (e.g., persistent procurement difficulties or route characteristics) and proactively mitigate or buffer them in current-period decisions (e.g., preemptively increasing budgets, adjusting procurement sources, or rerouting transportation). This effectively reduces the probability of violating hard constraints. Therefore, increasing the historical dependency depth can enhance the robustness and safety of dynamic decisions, thereby ensuring the smooth execution of resource allocation.

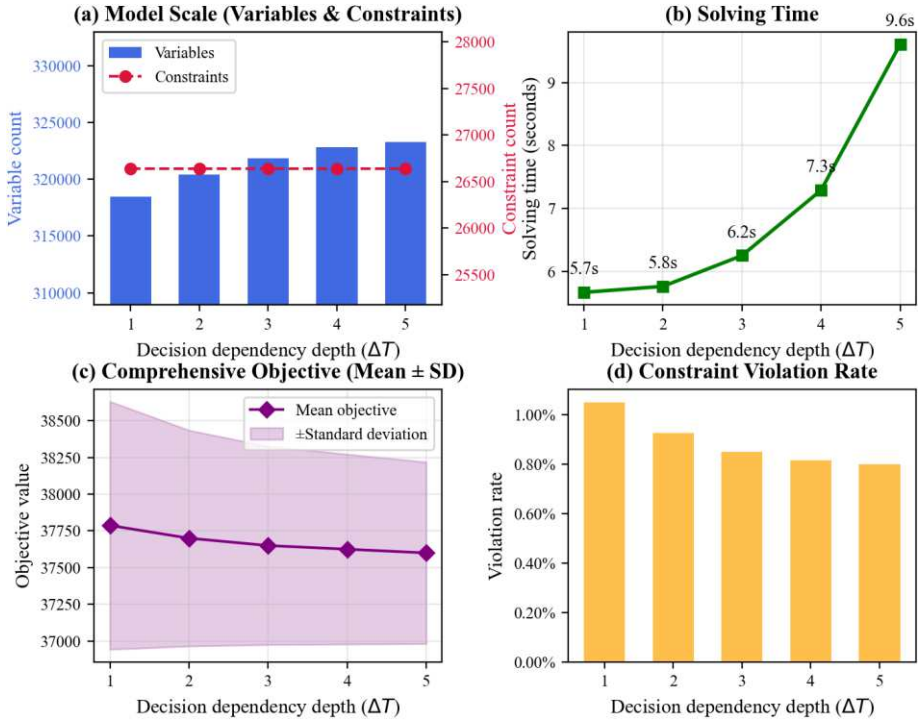

Figure 15 Comprehensive performance metrics under different decision dependency depths (ΔT)

6.4 Comparative Advantages of the Proposed Model

To comprehensively evaluate the performance advantages of the proposed multi-agent collaborative dynamic allocation model, this section conducts a comparative analysis with three representative methods across multiple performance metrics. **1) Deterministic Model (DM)** ignores all uncertainties (i.e., the model from Section 3), performing a single optimization based solely on pre-disaster estimates of parameters to generate a fixed resource allocation scheme. **2) Static Robust Optimization (SRO)** employs single-stage robust optimization (i.e., Δt=0), generating a conservative scheme feasible for the worst-case scenario within the predefined uncertainty set (Eq. (17)) for the entire rescue period. This scheme remains unchanged once formulated. **3) Rolling Horizon Optimization (RHO)** represents the current mainstream dynamic optimization approach. At the start of each decision period t, it resolves an optimization problem (referencing Section 3) based on the latest available information update and a limited look-ahead horizon, executing only the decisions for the current period t.

Using the identical case setup (T=6) from Section 6.1, the above comparative methods were simulated to calculate key metrics. For model scale and computational efficiency (**Figure 16** (a) and (b)), the scale of the proposed model and SRO is significantly larger than DM and RHO, leading to much longer solution times. This primarily stems from This primarily stems from introducing a number of auxiliary variables ($\boldsymbol{\beta}_t, \boldsymbol{\alpha}_1$) and constraints for robust dual transformation. In addition, MSARO additionally introduces slope variables ($\boldsymbol{Y}_t^1$) for the affine decision rules, which is a necessary cost for achieving multi-period dynamic adaptability and strict robust feasibility. DM, as a simple linear programming, is solved fastest. While each individual RHO solved within a rolling window has a model scale comparable to DM, the cumulative computational burden across multiple optimization runs over the entire rescue horizon is significantly higher than DM.

Regarding optimization performance (**Figure 16** (c)), DM, while theoretically yielding optimal allocation under its deterministic assumptions, suffers severe deviations in resource supply-demand gaps during actual execution due to ignoring perturbations. This leads to the highest (worst) comprehensive objective value f, which is unacceptable for affected populations. SRO sacrifices optimality in the comprehensive objective for worst-case feasibility, resulting in overly conservative solutions. RHO partially adapts to uncertainty through periodic re-optimization but suffers from myopic decision-making due to its independent solution per window, thereby often converging to local optima and limiting its optimization potential. In contrast, the proposed model dynamically adjusts current and future decisions based directly on historically disclosed uncertainty information. This enables not only reactive adaptation but also proactive, predictive mitigation of future impacts caused by historical uncertainties. Crucially, it coordinates fund, procurement, and transportation flows at the global level across the entire rescue period. The model explicitly considers interdependencies between allocation strategies across periods and the propagation of uncertainty over time during optimization. This ensures that the generated scheme is globally near-optimal from a multi-period perspective, thereby significantly improving the optimality (separately ~50%, ~44% and ~26% relative to DM, SRO, and RHO) and stability of the comprehensive

objective.

In the aspect of constraint reliability (**Figure 16** (d)), DM, ignoring uncertainty, exhibits the highest constraint violation rate, i.e. ~99%. This indicates DM frequently leads to fund chain breaks or resource inventory depletion, thereby rendering the solution practically infeasible. SRO, designed for strict feasibility within its predefined uncertainty set, achieves a low violation rate, i.e. ~3%, but its conservatism severely compromises optimization performance. RHO's violation rate (~35%) is significantly higher than SRO's. Its key weakness lies in insufficient cross-period coordination: decisions made in one period based on current information may lead to subsequent periods violating core constraints when encountering unanticipated severe uncertainties later. Its independent solutions struggle to effectively mitigate this intertemporal risk. Conversely, the proposed model demonstrates exceptional decision reliability with the lowest violation rate (0.8%). It is not only strictly constructed based on the predefined uncertainty set but also explicitly models the dependence of decision variables on historical uncertainties. This ensures current-period decisions satisfy current constraints while simultaneously providing flexibility and buffer space (via feedback adjustment rates) to handle future uncertainties, substantially avoiding the intertemporal violation risks inherent in RHO.

In summary, while the computational efficiency of the proposed model is not its primary advantage, it generates a globally optimized scheme covering the entire rescue period with dynamic adaptability. More importantly, this only requires acceptable solving times (seconds level), meeting the timeliness requirements of emergency response. Specifically, the proposed model overcomes the conservatism of SRO and the local optimality issues of RHO, thereby demonstrating superior optimization performance and stability. It means this study significantly reduces emergency economic costs and resource supply-demand gaps while enhancing rescue utility. Furthermore, it provides exceptionally high decision reliability, offering crucial safety guarantees for the sustainability of fund chains and resource inventories during emergency decision-making.

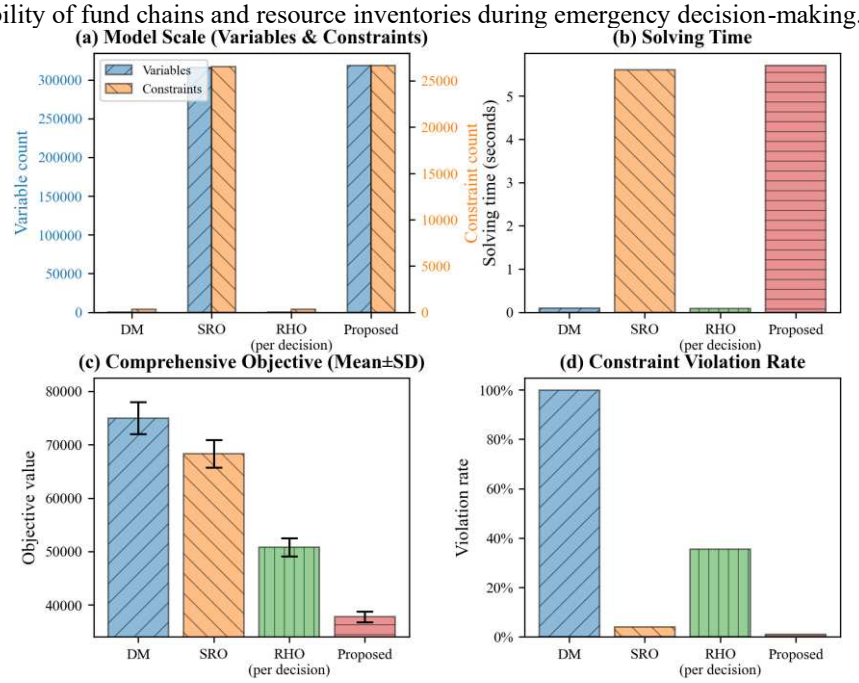

Figure 16 Performance metrics comparison under different scheduling models

7 Conclusions

Efficient and collaborative emergency resource allocation during the critical post-disaster "golden 72-hour window" is paramount for mitigating disaster losses. Addressing two critical bottlenecks in existing research—the lack of multi-agent coordination mechanisms and insufficient dynamic adaptability—this paper proposes an allocation method integrating a tripartite "Decision-Execution-Demand" linkage mechanism. This method synergizes pre-disaster planning with post-disaster feedback adjustments, achieving comprehensive optimization of economic costs, rescue utility, and supply-demand gaps. It provides an effective strategy for emergency resource allocation in complex disasters. Case study results based on an earthquake scenario demonstrate:

- **Effectiveness of the Allocation Strategy:** The tripartite linkage mechanism drives closed-loop feedback among fund allocation (decision agent), resource procurement (execution agent), and resource transportation (execution agent→demand agent). Under multiple uncertainty disturbances, the model almost ensures the security of the fund chain and resource inventory buffers through multi-period feedback adjustments to the pre-disaster plan. Crucially, it significantly enhances rescue efficacy and resource supply-demand balance while controlling economic costs.

- **Impact of Key Parameters on Model Performance:** The number of time periods (T) and the decision dependency depth (Δt) jointly influence model performance. A benefit-cost equilibrium point is reached at T=6: the comprehensive optimization objective decreases by ~5%, the constraint violation rate drops to 1%, and the solution time is controlled within 10 seconds. Excessively fine time granularity (T>9) leads to drastically increased computational burden with diminishing marginal optimization gains. Extending Δt can also improve objective optimization and decision safety at a relatively low computational cost.
- **Outstanding Comprehensive Performance:** Compared to the baseline DM, traditional SRO, and mainstream RHO, the proposed model achieves the best (lowest and most stable) comprehensive objective value, representing a substantial improvement of ~50%. This superior performance benefits economic cost, rescue utility, and resource supply-demand gap. Its constraint violation rate (0.8%) is 98% and 34% lower than DM and RHO, respectively, indicating significantly enhanced decision reliability. The model effectively overcomes the vulnerability of DM, the conservatism of SRO, and the local optimality issues of RHO.

This study provides a theoretical model for disaster emergency resource allocation characterized by coordination, adaptability, and reliability. The framework is extensible to managing sudden crises like public health events and major accidents. On this basis, future research can focus on: 1) Developing data-driven uncertainty sets and nonlinear decision rules to enhance dynamic adaptability. 2) Integrating disaster evolution prediction models to enable proactive cross-period optimization. 3) Designing coordination mechanisms and game theory-based agent incentive strategies for multi-disaster chain scenarios.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

1 Huang, Y. *et al.* Trends and Evolution in the Study of China's Natural Disaster Emergency Plans: A Bibliometric Analysis. *Evidence in Earth Science* **1**, 33-51 (2025).

2 Zhou, B. & Qian, J. Changes of weather and climate extremes in the IPCC AR6. *Advances in Climate Change Research* **17**, 713 (2021).

3 China, M. o. E. M. o. t. P. s. R. o. (ed Reduction and Relief Committee The Office of the National Disaster Prevention, the Ministry of Emergency Management) (2025).

4 Municipality, E. M. B. o. S. (ed Emergency Management Bureau of Shenzhen Municipality) (2019).

5 Wang, S. & Sun, B. Model of multi-period emergency material allocation for large- scale sudden natural disasters in humanitarian logistics: Efficiency, effectiveness and equity. *International Journal of Disaster Risk Reduction* **85**, doi:10.1016/j.ijdrr.2023.103530 (2023).

6 Bi, X. Q. & Zhang, D. in *4th International Conference on Mechanical Materials and Manufacturing Engineering (MMME).* 694-696 (2016).

7 Wang, S. R. *Optimization of location-routing problem on post-earthquake emergency logistics systems*, Zhejiang University, (2010).

8 Farahani, R. Z., Lotfi, M. M., Baghaian, A., Ruiz, R. & Rezapour, S. Mass casualty management in disaster scene: A systematic review of OR&MS research in humanitarian operations. *European Journal of Operational Research* **287**, 787-819, doi:10.1016/j.ejor.2020.03.005 (2020).

9 Wu, D., Yu, Q., Huang, M. & Liu, L. in *5th International Conference on Fuzzy Systems and Knowledge Discovery.* 617-620 (2008).

10 Yi, W. & Ozdamar, L. A dynamic logistics coordination model for evacuation and support in disaster response activities. *European Journal of Operational Research* **179**, 1177-1193, doi:10.1016/j.ejor.2005.03.077 (2007).

11 Davis, J. S., Allan, B. J., Pearlman, A. M., Carvajal, D. P. & Schulman, C. I. Optimal emergency personnel allocation after a natural disaster. *American journal of disaster medicine* **7**, 31-36 (2012).

12 Li, W. *et al.* A new two-stage emergency material distribution framework for urban rainstorm and flood disasters to promote the SDGs. *Sustainable Cities and Society* **112**, doi:10.1016/j.scs.2024.105645 (2024).

13 Procter, E. *et al.* Search and Rescue Response to a Large-Scale Rockfall Disaster. *Wilderness & Environmental Medicine* **26**, 68-71, doi:10.1016/j.wem.2014.07.007 (2015).

14 Ren, Z. & Zhang, Q. Volunteer dispatch considering fatigue effect and satisfaction in emergency situation. *Discover Applied Sciences* **6**, doi:10.1007/s42452-024-05667-x (2024).

15 Wu, Y. *et al.* A Dynamic Game Model for Emergency Resource Managers and Compound Disasters Induced by Heavy Rainstorms. *Water* **16**, doi:10.3390/w16202959 (2024).

16 Wu, J., Wang, Z., Song, Y., Zhou, X. & Cheng, L. A reinforcement learning-assisted search and rescue resource allocation decision-making approach for maritime emergencies. *Computers & Industrial Engineering* **201**, doi:10.1016/j.cie.2025.110933 (2025).

17 Zheng, Y. *Cooperative and Resilient Energy Compensation in the Smart Grid Using Mobile-Aware Electric Vehicles*, Chongqing University, (2024).

18 Yang, Q., Wang, F., Lu, J., Wang, Z. & Ma, B. Constrained vehicle routing model for emergency relief supplies based on demand urgency. *Journal of Tsinghua University (Science and Technology)* **64**, 1082-1088 (2024).

19 Fang, J. *et al.* Data Fusion in Forecasting Medical Demands based on Spectrum of Post-Earthquake Diseases. *Journal of Industrial Information Integration* **24**, doi:10.1016/j.jii.2021.100235 (2021).

20 Wu, X., Zhang, Y., Xiong, M., Mao, B. & Na, Z. Multi-carrier based UAV data relaying and user tracking system. *Physical Communication* **49**, doi:10.1016/j.phycom.2021.101462 (2021).

21 Cheng, J., Gao, Y., Tian, Y. & Liu, H. GA-LNS Optimization for Helicopter Rescue Dispatch. *Ieee Transactions on Intelligent Vehicles* **8**, 3898-3912, doi:10.1109/tiv.2023.3272621 (2023).

22 Tian, T., Wu, H., Wei, H., Wu, F. & Xu, M. An Efficient Route Planning Algorithm for Special Vehicles with Large-Scale Road Network Data. *Isprs International Journal of Geo-Information* **14**, doi:10.3390/ijgi14020071 (2025).

23 Yang, J., Hou, H., Qin, B. & Destech Publicat, I. in *International Conference on Modern Education and Information Technology (MEIT).* 459-463 (2017).

24 Liu, J. *et al.* Medical Rescue and Treatment during Two Weeks after Min-Zhang Earthquake in Gansu Province. *Chinese Journal of Evidence-Based Medicine* **13**, 906-911 (2013).

25 China, C. P. s. G. o. t. P. s. R. o. (ed Central People's Government of the People's Republic of China) (2024).

26 Su, Z. P., Zhang, G. F., Liu, Y., Yue, F. & Jiang, J. G. Multiple emergency resource allocation for concurrent incidents in natural disasters. *International Journal of Disaster Risk Reduction* **17**, 199-212, doi:10.1016/j.ijdrr.2016.05.003 (2016).

27 Xuelong, C., Zirui, W. & Miaomiao, L. Research on the conflict resolution method of multi-period emergency resource allocation subjects. *Chinese Journal of Management Science*, 1-22 (2024).

28 Shariatmadar, K. & Versteyhe, M. in *2019 7th International Conference on Control, Mechatronics and Automation (ICCMA).* 84-89 (IEEE).

29 Qin, J., Zou, Z., Liu, L. & Liu, Y. Piecewise affine decision rules for contextual chance-constrained stochastic programming. *Operations Research Letters* **61**, 107296 (2025).

30 Zhang, L. *et al.* Enhancing robustness: Multi-stage adaptive robust scheduling of oxygen systems in steel enterprises under demand uncertainty. *Applied Energy* **359**, doi:10.1016/j.apenergy.2024.122729 (2024).

31 Lappas, N. H. & Gounaris, C. E. Multi-stage adjustable robust optimization for process scheduling under uncertainty. *AIChE Journal* **62**, 1646-1667 (2016).

32 Xu, G. & Hanasusanto, G. A. Improved decision rule approximations for multistage robust optimization via copositive programming. *Operations Research* **73**, 842-861 (2025).

33 Jun, C. *et al.* Wenchuan Earthquake Disaster Atlas Compilation Project. (2009).

34 China, T. C. o. t. P. s. R. o. 2020 National Toll Road Statistical Bulletin. (2021).

35 Jianwei, C., Weiguo, W., Youpo, S. & Xin, L. Research on earthquake emergency rescue resource allocation model. *World Earthquake Engineering* **29**, 33-37 (2013).

36 Zhu, Y., Tian, D. & Yan, F. Effectiveness of entropy weight method in decision-making. *Mathematical problems in Engineering* **2020**, 3564835 (2020).

37 Lin, Z., Mengtao, L. & Jun, T. Research on physical and raw material reserve strategies for emergency material supply agreement companies considering post-disaster spot market procurement. *Systems Engineering - Theory & Practice* **42**, 1001 (2022).